

Milestone - 2

Model Training Report

Dataset Used:

VinBigData Chest X-ray Abnormalities Detection



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1. Introduction

CliniScan is an AI-powered system designed to assist radiologists in detecting and classifying abnormalities in chest X-rays. Milestone 2 focuses on training two deep learning models:

- A **Classification Model** for disease presence prediction
- A **Detection Model** for bounding box localization

This milestone aims to establish baseline performance for both tasks using a subset of the dataset.

2. Objective of Milestone 2

The key objectives were:

- Train a multi-label classification model
- Train an object detection model
- Evaluate both models using appropriate metrics
- Document model performance
- Save trained models for future optimization

3. Dataset Description

The dataset contains chest X-ray images labeled with:

- 14 abnormality classes
- Bounding box annotations
- “No Finding” cases

A subset of **5000 images** was selected for training and validation.

Key Dataset Characteristics:

- Image format: PNG (converted from DICOM)
- Resolution standardized to 512×512
- YOLO format annotations for detection
- Multi-label format for classification

4. Data Verification and Preprocessing

Before training, several checks were performed:

4.1 Image Size Verification

All images were resized to 512×512 for consistency.

4.2 Label Matching

Each image was verified to have a corresponding annotation file.

4.3 Missing Annotation Check

No missing bounding box files were found.

4.4 Dataset Split

Dataset was split into:

- 80% Training
- 20% Validation

Directory structure:

images/train

images/val

labels/train

labels/val

4.5 Bounding Box Visualization

Random samples were visualized with bounding boxes to ensure annotation correctness.

5. Classification Model Development

5.1 Dataset Preparation

For classification, bounding boxes were converted into image-level labels.

If an abnormality appeared in an image, that class was marked as 1.

This resulted in a **multi-label classification dataset**.

5.2 Model Architecture

The selected architecture:

ResNet-50

Reasons:

- Strong baseline performance
- Pretrained on ImageNet
- Good transfer learning capability

Modifications:

- Final fully connected layer replaced with 14 output neurons
- Sigmoid activation used for multi-label classification

5.3 Training Configuration

Parameter	Value
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Optimizer	Adam
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Learning Rate	0.001
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Loss Function	BCEWithLogitsLoss
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Epochs	25
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Batch Size	16
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Image Size	512×512
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Data augmentation applied:

- Horizontal flip
- Minor brightness variation

5.4 Validation Strategy

Validation metrics used:

- AUC-ROC
- F1-Score
- Validation Loss

The best model was selected based on **highest validation F1-score**.

Model saved as:

best_classification_model.hcl

5.5 Classification Results

Observed Baseline Performance:

- Average AUC: 0.81
- Macro F1 Score: 0.68
- Validation Loss reduced steadily

The model demonstrated strong ability to detect common abnormalities such as:

- Opacity
- Consolidation
- Pleural Effusion

However, rare classes showed lower F1 scores due to class imbalance.

6. Detection Model Development

6.1 Annotation Conversion

Annotations were converted to YOLO format:

class_id x_center y_center width height

Coordinates normalized between 0 and 1.

6.2 Model Selection

Selected detection model:

YOLOv8

Reasons:

- State-of-the-art performance
- Fast training
- Built-in augmentation
- Easy evaluation pipeline

Nano version (yolov8n) used for baseline.

6.3 Training Setup

Parameter	Value
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Epochs	50
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Batch Size	16
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Image Size	512
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Learning Rate	0.001
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Optimizer	SGD (default YOLO)
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Safe medical augmentations applied:

- Horizontal flip
- Scaling
- Mild rotation

6.4 Evaluation Metrics

Detection evaluated using:

- mAP@0.5
- mAP@0.5:0.95
- Precision
- Recall
- Box Loss

6.5 Detection Results

Baseline Results:

- mAP@0.5: 0.42
- mAP@0.5:0.95: 0.27
- Precision: 0.64
- Recall: 0.58

Commonly detected abnormalities:

- Lung Opacity
- Consolidation
- Cardiomegaly

Small objects such as nodules had lower detection accuracy.

Best model saved as:

best.pt

7. Performance Analysis

Strengths

- Strong baseline performance with limited data
- Good generalization on validation set
- Stable loss convergence

Weaknesses

- Class imbalance affected rare disease detection
- Small bounding boxes harder to detect
- Some false positives in overlapping regions

8. Challenges and Solutions

Challenge	Solution
Class imbalance	Used weighted loss for classification
Large DICOM files	Converted to PNG
Annotation inconsistency	Verified via visualization
GPU memory limits	Reduced batch size

9. Conclusion

Milestone 2 successfully achieved:

- Development of a classification model
- Development of an object detection model
- Establishment of baseline metrics
- Proper documentation and saved models

The detection model achieved moderate mAP scores, while the classification model demonstrated strong disease presence prediction capability.

Both models provide a solid foundation for further optimization in Milestone 3.
